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Floppy, Hard-Disk and Stick Storage of the G2 Amiga

Every drive of the Amiga is a file on one USB stick: four floppy drives from ADF images, two Gayle IDE hard disks from HDF images with an 8 MB write-back cache, and a stick folder as the AmigaOS volume USBStick:, served by the RISC-V firmware through the core's sector port.

HIGHLIGHTS

- DF0:–DF3: from DD ADF images, held in DDR2.
- HD0: and HD1: from HDF images; new images 128 MB–1 GB made from setup.
- 8 MB write-back cache of 64 KB blocks; files synchronised 1 s after the last write.
- USBStick: volume on a stick folder.
- FAT32 and exFAT sticks.

KEY PARAMETERS

PARAMETER	VALUE
Floppy drives	4
Floppy image	901,120 bytes (DD)
Hard disks	2 (IDE master, slave)
New HDF	128 MB–1 GB
HD cache	128 × 64 KB
HD read / write	609 / 620 KB/s
Stick FS	FAT, exFAT

1 THE STICK

ITEM	VALUE
File system	FAT12, FAT16, FAT32 or exFAT; long names; FatFs R0.16 (ChaN)
Directory	\amiga: Kickstart images, AMIGA.INI, disk images, AMIGA.FW, BOOT.LOG. Nothing outside \amiga is read or written, except the SHARE: folder
Sticks tested	USB 2.0 high-speed sticks; a stick that is not FAT/exFAT formatted is not mounted

2 FLOPPY DRIVES DF0:–DF3:

ITEM	VALUE
Images	*.adf of 901,120 bytes (DD: 80 cylinders × 2 heads × 11 sectors). An HD-size image (1,802,240 bytes) is accepted by the firmware's tools, but AmigaOS reads it as not a DOS disk (test unit)
Controller	Paula's floppy controller in the core; sectors served by the firmware
Buffering	Each image read whole into DDR2 chip A (2 MB a drive) when inserted; every request is answered from there
Writes	A written sector goes to the copy and to the file at once; the file is synchronised 1 s after the last write. WRITE=0 protects every drive
Change	From setup (Drives page) or AMIGA.INI DF0=–DF3=; a change is a disk change for trackdisk.device
Measured	A 204,800-byte file copied by AmigaOS to a new FFS image reads back with an equal sha256

3 HARD DISKS HD0: AND HD1: (GAYLE IDE)

ITEM	VALUE
Images	*.hdf with a RigidDiskBlock; the core takes the geometry from it. Gayle IDE master (HD0) and slave (HD1)
New image	From setup: NEW_<size>M.hdf, 128 MB to 1 GB, RigidDiskBlock at block 0 (16 heads, 63 sectors a track), one bootable partition DH<n> of DosType DOS\3 from cylinder 2; Workbench formats it
Cache	Write-back, 128 blocks of 64 KB (8 MB) in DDR2 chip A shared by both drives, least recently used replaced; a changed block goes to the file in one 64 KB write when replaced, 1 s after the last write, after a console file command and before the file is closed
Rate through scsi.device	32 KB reads 609 KB/s, writes 620 KB/s; a 512-byte block about 1.2 ms
Measured	AmigaOS writes to HD0: 0 of 400 sectors lost, 0 timeouts

4 USBSTICK: (THE SHARE: VOLUME)

ITEM	VALUE
Function	A directory of the stick (AMIGA.INI SHARE=, default /amiga/share, made when missing) is the AmigaOS volume USBStick of the device SHARE:
Amiga side	L:share-handler with DEVS:DOSDrivers/SHARE (from a314fs by Niklas Ekström, CC0-1.0)
Transport	Blocks 4–143 of HD0, answered by the firmware and never written to the image: requests, responses and 64 KB of file data
Names	Case ignored; code page 437 on the stick, Latin-1 on the Amiga (characters above 127 untested)
Not stored	Protection bits and comments (FAT has no field for them); SET_PROTECT and SET_COMMENT succeed
Limits	48 locks, 8 open files; a file in use as a drive image is not opened, deleted or renamed

5 FIRMWARE WRITES INTO IMAGES

ITEM	VALUE
afs	Console verbs info, ls, stat, get, put, mkdir, mv, rm, untar, format, hold and release, inside the Amiga volumes of DF0–DF3 and HD0–HD1, on ADFlib 0.10.7 (LGPL-2.1)
Hard disks	PANOFS (C:PANOFS, started from S:User-Startup) inhibits the volume through AmigaDOS while the firmware writes, then releases it; without PANOFS the firmware stops the core, writes and cold-starts the machine
Floppies	Ejected from the core, written, inserted again
Measured	afs untar: 10,840,064 bytes, 937 files, 137 directories in about 600 s

6 RESERVED BLOCKS OF HD0

BLOCKS	USE	CONDITION
2	PANOFS mailbox	The RigidDiskBlock uses blocks 0–1
3	Command block: a console command from AmigaOS and its output	Blocks 0–2
4–143	SHARE: requests and file data	Blocks 0–3, partitions after 143
144, 145, 256–383	Configuration-flash window (panoflash.device)	Partitions at 384 or later
511–895	Socket window (PanoTCP)	Partitions at 896 or later

Images made by setup start their partition at block 2,016, so every window is available. An image partitioned elsewhere with blocks in these ranges keeps them: the firmware answers a window only when the RigidDiskBlock leaves it free.

7 LIMITS AND UNTESTED ITEMS

ITEM	STATE
Images above 1 GB, two partitions	Untested
Stick removal while the machine runs	Not supported: the drives lose their files; power-cycle the unit
Names above character 127 on USBStick:	Untested

Measured on a G2 revision C unit, firmware release 1.0. Repository mikewolak/pano-g2-amiga, release 1.0.